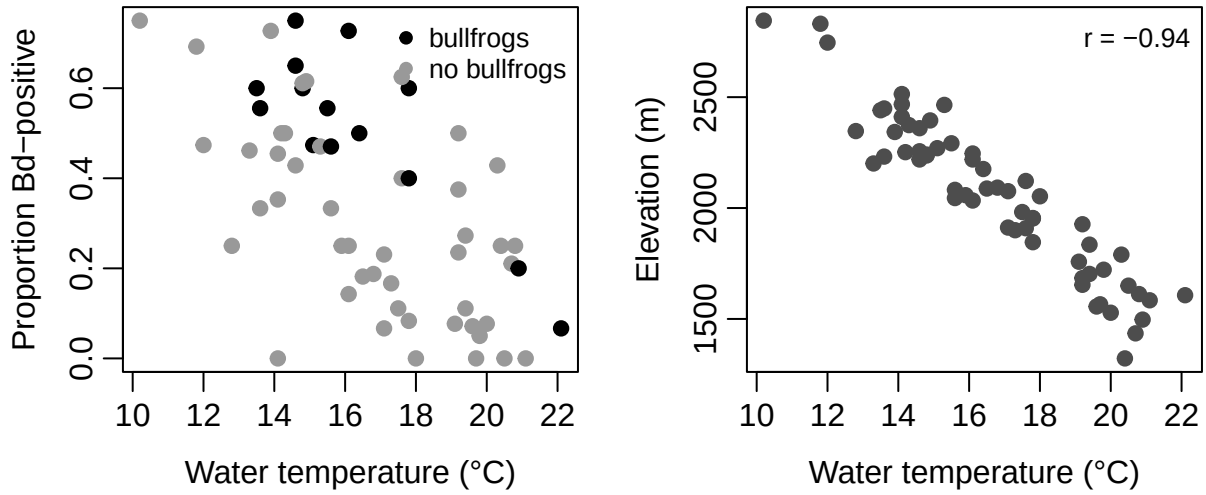
Lab 8, Act 1: Which Pond Features Matter?

Which pond features predict chytrid, and can the data tell temperature from elevation?

Answer key. Answers are in magenta sans-serif type; longer answers have a bar in the margin.

The scenario

The summer after the 12-pond survey from Lab 7, the field crew swabbed frogs at **60 ponds** across the region and recorded eight features of each pond: water temperature, elevation, canopy cover, conductivity, pond area, whether fish are present, whether bullfrogs are present (an invasive species that carries Bd without dying from it), and distance to the nearest road. Managers can afford to monitor only some ponds and want to know which features predict where Bd is common.



The model for every candidate is the logistic-binomial model from Lab 7, with more predictors: the number of Bd-positive frogs at pond i is Binomial(n_i, p_i), with

$$\text{logit}(p_i) = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots$$

Every predictor was **standardized** (mean 0, SD 1) before fitting, so each β is the change in log-odds of infection per standard deviation of that predictor.

Which pond features predict Bd prevalence?

Please keep laptops closed for this Act.

Part 1: Comparing candidate models (individual, ~20 min)

The table lists seven candidate models, the number of parameters k (including β_0), and each model's negative log-likelihood (NLL) at its MLE.

1a. Fill in the missing AIC values ($\text{AIC} = 2\,\text{NLL} + 2k$) and every ΔAIC.

Model	Predictors	k	NLL	AIC	dAIC
1	intercept only	1	179.40	360.80	*139.64*
2	temp	2	137.42	*278.84*	*57.68*
3	elevation	2	137.73	*279.46*	*58.30*
4	temp + elevation	3	136.39	278.78	*57.62*
5	temp + bullfrogs + canopy	4	106.58	*221.16*	*0.00*
6	temp + elev. + bullfrogs + canopy	5	106.20	222.40	*1.24*
7	all eight	9	104.59	*227.18*	*6.02*

1b. Which model is best by AIC? Model 6 adds elevation to Model 5 and sits within 2 AIC units of it. Does that mean the data support an effect of elevation? Does the difference in the two models' raw NLLs help you answer this question, and if so, how?

> Model 5 (temp + bullfrogs + canopy), AIC 221.16. No. Yes, the NLLs help: adding elevation lowers the NLL by only 0.38, less than the 1 NLL unit (2 AIC units) an extra parameter must buy to lower AIC. A parameter with no effect at all costs at most 2 AIC units, so landing within 2 is expected even if elevation does nothing.

1c. Use likelihood ratio tests to compare Model 5 with Model 6, and Model 5 with Model 7. For each, give the test statistic, its degrees of freedom, and your conclusion. Critical values of χ^2 at the 0.05 level: 3.84 (1 df), 5.99 (2), 7.81 (3), 9.49 (4), 11.07 (5).

> Model 5 vs. 6: $D = 2(106.58 - 106.20) = 0.76$, 1 df; $0.76 < 3.84$, so keep Model 5.
>
> Model 5 vs. 7: $D = 2(106.58 - 104.59) = 3.98$, 5 df; $3.98 < 11.07$, so keep Model 5.

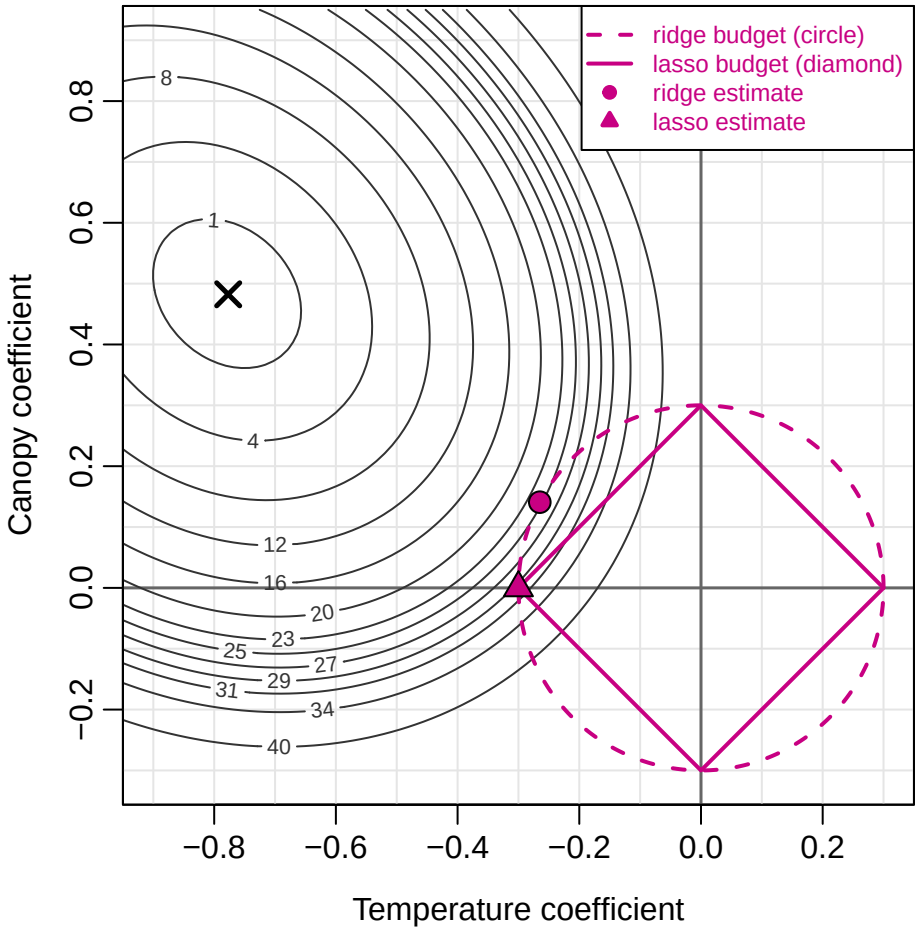
1d. Models 2 and 3 are nearly tied. Can these data tell whether water temperature or elevation is what drives Bd? Why or why not? What kind of ponds, or what kind of study, would help?

> No. Temperature and elevation are strongly correlated ($r = -0.94$), so either one alone fits about equally well (ΔAIC 0.62). Helpful: ponds where the two come apart (warm high ponds, cold low ponds, e.g. spring-fed or shaded sites), or an experiment that changes water temperature directly.

Part 2: Penalties as budgets (individual, ~15 min)

A ridge penalty, $\text{NLL}(\beta) + \lambda \sum_j \beta_j^2$, gives the same estimate as finding the lowest-NLL point inside a **budget** $\sum_j \beta_j^2 \le t^2$, for some t that shrinks as λ grows. The lasso budget is $\sum_j |\beta_j| \le t$. Neither budget includes β_0.

The figure shows NLL contours for a model with only two predictors, water temperature and canopy cover, labeled by how far each lies above the minimum. The × marks the MLE.



2a. On the figure, draw the ridge budget $\beta_{temp}^2 + \beta_{canopy}^2 \le 0.3^2$ and the lasso budget $|\beta_{temp}| + |\beta_{canopy}| \le 0.3$. Mark the ridge estimate and the lasso estimate, and read off their approximate coordinates.

Ridge: (-0.26, 0.14) Lasso: (-0.30, 0.00) (approximate values from a sketch are fine)

2b. Did either penalty set a coefficient to exactly 0? Could a ridge budget ever set a coefficient to exactly 0? Use the shapes in your sketch to explain.

> The lasso did: its estimate sits on the diamond's corner, $(-0.30, 0.00)$, so the canopy coefficient is exactly 0. Ridge did not (canopy 0.14). Ridge essentially never does: the circle has no corners, so the lowest contour touches it on an axis only if the MLE already has that coefficient at exactly 0.

2c. In general, when does the lasso set a coefficient to exactly 0? Describe what the contours and the budget have to look like.

> When the lowest contour reaches the diamond first at a corner (on an axis) rather than along an edge. That happens when the budget is small compared with how far the MLE is from 0, and the contours approach the diamond from a corner's direction (one coefficient's effect is weak relative to the other's). A smaller budget zeros more coefficients. Contours stretched parallel to an edge (strongly correlated predictors) tend to touch the edge instead, keeping both.

Group discussion and revision (~10 min)

Compare your answers with your group, especially 1b, 1d, and your sketch in 2a, and revise as needed. Record any changes you made, new insights, and any "muddiest points" or topics you want to review.

Open-ended.